Aaron W. Storey

AI Research Engineer | Trustworthy AI Evaluation
Model Selection for Language-Based Decisions | Responsible AI

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Research Profile

AI research engineer and PhD candidate. Engineering work includes multi-agent research prototypes with traceable evidence and reproducible evaluation; applied work spans biometrics and GNSS-denied navigation. Earlier career: 15 years of Army service.

Research focus Deciding when a language classifier is adequate for a specific decision and whether extra inference computation earns its cost, testing candidate models against requirements from published deployment practice with explicit uncertainty.

Research interests Model evaluation and selection; uncertainty quantification and calibration; NLP classification; human-in-the-loop decision support; responsible AI; agentic systems and verification.

Education

Jan 2025 to Present — **Ph.D. Candidate, Computer Science**, *Clarkson University*, Potsdam, NY
Cumulative GPA **3.98/4.00**. Advisor: Jeanna Matthews. Member, Explainability & Transparency Group, since Sep 2026.

Aug 2023 to Dec 2024 — **M.S., Artificial Intelligence**, *Maryville University of Saint Louis*
Cumulative GPA **3.97/4.00**. Conferred December 2024.

2017 — **B.S., Information Technology**, *American Military University*
Completed during Army service.

Publications

Publication statuses current as of September 8, 2026.

Published

2026 Kia, A.; **Storey, A. W.**; Imtiaz, M. "Advanced Hardware Security on Embedded Processors: A 2026 Systematic Review." *Electronics*, 15(5), 1135, 2026. Published March 9, 2026. doi:10.3390/electronics15051135

Peer-Reviewed Conference Papers

2026 **Storey, A. W.**; Imtiaz, M. "A Three-Channel Audit of the Biometric Evidence Chain." *4th IEEE International Conference on Artificial Intelligence, Blockchain, and Internet of Things (AIBThings 2026)*. Presented September 6, 2026. **Accepted; proceedings publication pending.**

2026 **Storey, A. W.**; Kia, A.; Imtiaz, M. "LLM on the Edge: A 2026 Perspective." *4th IEEE International Conference on Artificial Intelligence, Blockchain, and Internet of Things (AIBThings 2026)*. Presented September 6, 2026. **Accepted; proceedings publication pending.**

Accepted; Presentation Pending

2026 **Storey, A. W.**; Ahmed, A.; Sukhdeve, Y.; Imtiaz, M. H.; Mondal, S. "Nonparametric Template Persistence Curves for Child Face Recognition via Beta Generalized Additive Models." *IEEE UEMCON 2026*. Accepted and presented October 11, 2026.

Research and Engineering Appointments

Aug 2025 to Present — **Independent AI Research Engineer**, *Kymera Systems LLC*, San Antonio, TX

- Develop agentic and multi-agent research prototypes combining heterogeneous models, structured tool use, and evaluation workflows.
- Design verification workflows that track intermediate artifacts, provenance, persistent state, and the evidence supporting generated conclusions.
- Build experimental agent-memory and knowledge-representation systems, local AI infrastructure, and research software as an individual contributor.

Jun 2026 to Sep 2026 — **Graduate Research Assistant, Autonomous Systems and AI**, *Clarkson University, AVHBAC Laboratory*, Remote

- Developed simulation and evaluation workflows for GNSS-denied navigation, state estimation, sensor fusion, and lunar surface perception using Python, ROS2, and OpenCV.
- Investigated visual odometry, pose-graph optimization, and terrain-referenced localization for autonomous systems operating in physically constrained environments.
- Contributed research software, manuscripts, seminars, and grant-development materials connecting perception, simulation, and autonomous decision-making.
- Released **ofiqpy** on PyPI under AVHBAC (v0.1.1, July 2026; v0.2.0, August 2026), implementing the 27 quality components exposed by BSI OFIQ v1.1.0 plus its unified quality score.

Jun 2025 to May 2026 — **Graduate Research Assistant, Trustworthy AI and Biometrics**, *Clarkson University, Center for Identification Technology Research (CITeR)*, Remote

- Conducted biometric evaluation and face-image-quality research involving ISO/IEC 29794-5-related components, attribution faithfulness, calibration, uncertainty, and data quality.
- Built reproducible Python workflows for counterfactual analysis, perturbation testing, statistical evaluation, and standards-oriented measurement.
- Contributed manuscripts, systematic reviews, experimental protocols, and research software. Research supported through CITeR under DHS Science and Technology Contract 70RSAT24CB0000022.

Professional Service

2026 — Reviewer, **IEEE AIBThings 2026**.

Research Support and Grant Development

Support — Biometric and face-image-quality research supported by the Center for Identification Technology Research under **DHS Science and Technology Contract 70RSAT24CB0000022**. Role: graduate researcher.

Development — **Federal lunar surface autonomy proposal packages, 2026:** lead drafter on four mechanisms (LuSTR, ESI dual-anonymous, NSTGRO, and STTR). Prepared technical narratives, budgets, data-management plans, inclusion plans, and compliance checklists. Status: **drafted and unsubmitted; not awarded funding**.

Development — **NCSU Laboratory for Analytic Sciences (LAS) 2027 Call for Abstracts:** co-authored abstract 7842, "Which Recognizer Hears Which Sounds? Phoneme and Language-Aware Routing for Degraded Multilingual Speech," with Y. Sukhdeve (M. H. Imtiaz, Principal Investigator). Selected in August 2026 with an invitation to submit a full project proposal, due September 30, 2026. Abstract stage; not awarded funding.

Research Software and Selected Technical Projects

circle

GitHub-native research collaboration framework in Go with shared memory, artifact tracking, review gates, context packs, and a git-native knowledge graph. An MCP server exposes 22 tools to AI assistants. Releases v0.1.0 to v0.3.0.

ofiqpy

Python implementation of the fixed BSI OFIQ v1.1.0 quality-assessment profile: 27 components plus the unified quality score, configuration and model integrity checks, and a live C++ comparison workflow. PyPI v0.2.0 released August 2026.

okf-tools

Python tools to validate, lint, migrate, and index Open Knowledge Format bundles, including detection of broken relationships and semantic failures not rejected by basic conformance checks.

lunar-terrain-sidecar

Lunar south-polar terrain generation, authoring, and preprocessing system for Godot-based robotics simulation. Uses LOLA/PGDA terrain products, ephemeris-derived illumination, explicit provenance, automated validation, and CI. Archived at doi:10.5281/zenodo.22014046.

swinvit-face-verification

Reproducible face-verification toolkit comparing SwinFace and ArcFace encoders through a common evaluation interface over public benchmark data, with documented provisioning, integrity checks, metrics, limitations, and optional mixed-effects analysis.

medicaid-kg

Knowledge-graph and geospatial exploration system for a 227-million-row HHS T-MSIS Medicaid provider-spending dataset, using DuckDB, NetworkX, FastAPI, React, MapLibre/deck.gl, and agent-assisted querying.

agent-keyring

Local credential broker for AI agents and programs on a single Linux host, built on Infisical agent-vault v0.40.0 with twenty local patches. Agents send placeholder keys through a loopback HTTPS proxy that attaches the real credential, so keys stay out of environment files, logs, and model context; a companion command-line tool runs agents in containers with network egress only through the proxy. Public snapshot, October 2026, with a written threat model and security audit.

hostdesk

Per-machine Linux service desk in Go that discovers systemd services, Docker and Podman containers, listening sockets, Tailscale routes, and projects, and offers guarded start, stop, and restart of permitted services. Release v0.1.0, October 2026, with amd64 and arm64 binaries and a checksum-verifying installer.

OpenFIQA Studio, *In development*

Local-first scientific IDE for biometric quality, verification research, controlled degradation experiments, reproducible evaluation, and publication-artifact generation. Design-stage work comprising product architecture, scientific schemas, and decision records.

Teaching and Educational Contributions

Summer 2025 **Guest Lecturer and Course Developer, EE622 Advanced Biometrics**, *Clarkson University*

Developed graduate-level course materials and delivered **nine guest lectures** covering advanced biometric systems and Transformer architectures across biometric modalities.

Dec 2023 to Jan 2024 **Co-Creator and Contributor, 100 Days of Machine Learning**

Co-created a free, 35-lesson curriculum progressing from Python foundations through applied machine learning and XGBoost; contributed instructional notebooks and supporting educational materials.

Invited Talks and Research Seminars

Jan 2026 **Behavioral Control Surfaces in Large Language Models.** Invited seminar, Department of Electrical and Computer Engineering, Clarkson University, January 22, 2026.

Jun to Jul 2026 **Navigating Worlds Without GPS: Off-World Autonomous Navigation.** Three-session public research seminar series, Clarkson University, June 18, June 25, and July 2, 2026. Topics included dead reckoning and SLAM, illumination as a navigation instrument, and autonomous construction on mutable terrain.

Prior Technical, Data, and Operational Experience

Jan 2018 to Jul 2023 **Career Transition**
Personal responsibilities and self-directed technical study preceding graduate education in artificial intelligence.

Aug 2002 to Jan 2018 **United States Army**
15 years of service across medical operations, personnel supervision, equipment accountability, technical training and simulation, readiness, information management, and deployed communications. Defense Health Agency roles below were Army assignments within this service period; their labels describe functions, not official duty titles.

Sep 2015 to Jan 2018 **Data and Operations Analyst**, *Army assignment, Defense Health Agency*, San Antonio, TX

- Analyzed operational data and information systems to support planning and decision-making; coordinated technology projects and supported information assurance, data integrity, and security requirements.

Jul 2013 to Sep 2015 **Educational Technology Specialist**, *Army assignment, Defense Health Agency*, San Antonio, TX

- Developed technology-enabled healthcare training and simulation scenarios and evaluated training performance with clinical, technical, and instructional teams.
- Evaluated 33 DMRTI and National Association of EMT TC3 instructors during the Jun 2014 to May 2015 evaluation period.

2012 to 2013 **Signal Plans and Operations**, *United States Army*, Kabul, Afghanistan

- Served as a technical advisor to military, civilian, and contractor personnel; led a U.S. systems team updating fiber-network infrastructure supporting U.S. and multinational communications.

2002 to 2012 **Earlier Army assignments.** Medical operations in Hawaii and Iraq; Ranger School medical support and training (2009 to 2011); V Corps medical logistics, readiness, and information-management duties in Germany (2011 to 2012). Responsibilities included emergency-response coordination, personnel supervision, and technical training. Maintained accountability for equipment valued at over $3.2 million with zero loss during the Jul 2008 to Jun 2009 evaluation period.

Mar to Jul 2009 **Web Architect and SEO Analyst**, *Possible Design, LLC*, Honolulu, HI
Designed and maintained web properties with responsibility for architecture, usability, performance, and search optimization. Managed web-development work from requirements through implementation and used analytics to guide technical improvements.

Feb to Apr 2009 **Web Architect and Data Analyst**, *IceBreakers Hawaii, LLC*, Honolulu, HI
Designed web architectures and analyzed site performance and engagement data. Translated organizational requirements into web-system functionality.

Technical Skills

Evaluation and Model Selection	Evaluation design; non-inferiority and equivalence testing; probability calibration (temperature and Platt scaling); selective prediction and deferral; interval estimation (Wilson bounds, bootstrap); transformer classifiers (ModernBERT, RoBERTa); LoRA fine-tuning; Hugging Face.
Agentic AI and Knowledge Systems	Multi-agent systems; agentic AI development; heterogeneous model orchestration; tool-using agents; persistent state; agent memory; provenance; knowledge representation; knowledge graphs; local and hybrid LLM workflows.
Verification, Evaluation, and Trustworthy AI	Verification and validation; AI reliability; AI safety; evidence quality; uncertainty; calibration; conformal prediction; explainable AI; attribution faithfulness; counterfactual and perturbation testing; preregistered ablations; benchmark design; reproducible evaluation; data-leakage audits.
Machine Learning and Computer Vision	PyTorch; scikit-learn; Vision Transformers; deep learning; neural networks; face recognition and verification; face-image quality assessment; image processing; visual odometry; stereo depth; image geolocation.
Autonomous Systems	ROS2; OpenCV; SLAM; state estimation; sensor fusion; SE(2)/SE(3) pose graphs; loop closure; terrain-referenced localization; GNSS-denied navigation; simulation; Godot.
Statistical and Scientific Computing	Python; R; SQL; Bash; beta regression; time-series analysis; statistical modeling; uncertainty quantification; confidence intervals; inter-rater agreement; Jupyter.
Systems and Data Engineering	FastAPI; DuckDB; Docker; Linux; Git; databases; data modeling; REST APIs; scientific software engineering; LaTeX.
Standards and Assurance	ISO/IEC 29794-5; ISO/IEC 19795; information assurance; standards-oriented testing; reproducibility and provenance controls.

Selected Recognition

Individual decorations	Joint Service Commendation Medal; Army Commendation Medal (5 awards); Joint Service Achievement Medal; Army Achievement Medal (5 awards).
Joint award	JBSA-Fort Sam Houston Star Performer Joint Award for leadership excellence in medical training.

Certifications and Professional Affiliations

CompTIA Security+, earned during military service. **IEEE Member**, since 2024.

Updated September 25, 2026.